

MSDM 5058 Information Science
Assignment 2 (due 14th March, 2026)

Submit your assignment solution on Canvas. You may discuss with others or seek help from your TA, but you should not directly copy from others. Otherwise, it will be considered as plagiarism.

(1) Average entropy

Let $H(p) = -p \log_2 p - (1 - p) \log_2 (1 - p)$ be the entropy function of a binary source.

- (a) Evaluate $H(1/3)$.
- (b) Calculate the expected entropy $H(p)$ when the probability p is chosen uniformly in the range $0 \leq p \leq 1$.

(2) Mutual information for correlated normal distributions

X and Y are two correlated random normal variables with the following joint probability distribution

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim N_2 \left(0, \begin{bmatrix} \sigma^2 & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 \end{bmatrix} \right)$$

Evaluate $I(X; Y)$ and comment on the cases when $\rho = 1$, 0 , and -1 .

(3) Channel capacity

Consider a binary asymmetric communication channel. Its input alphabet is $X = \{0, 1\}$ with probabilities $\{0.5, 0.5\}$, and its output alphabet is $Y = \{0, 1\}$. Its channel matrix is

$$\begin{pmatrix} 1 - \alpha & \beta \\ \alpha & 1 - \beta \end{pmatrix}$$

where α is the probability of transmission error when sending $X = 0$ and β is the probability of transmission error when sending $X = 1$.

- (a) What is the entropy $H(X)$ of the input?
- (b) What is the probability distribution $p(Y)$ of the outputs and the entropy $H(Y)$ of this output distribution?
- (c) What is the joint probability distribution $p(X, Y)$ of the input and the output? What is their joint entropy $H(X, Y)$?
- (d) What is the mutual information $I(X; Y)$ of this channel as a function of α and β ?
- (e) How many combinations of (α, β) are there for which the mutual information of

this channel is maximal? What are those values, and what then is the capacity of such a channel in bits?

- (f) What condition does (α, β) satisfy when the capacity of this channel is minimal? What is the channel capacity in that case?

(4) Shannon, Fano, and Huffman codes

A source X has an alphabet of seven characters $\{a, b, c, d, e, f, g\}$ with probabilities $\{0.01, 0.24, 0.05, 0.20, 0.47, 0.01, 0.02\}$.

- (a) What is the entropy of X ?
- (b) Give an example of a set of codewords for X using Shannon code, Fano code, and Huffman code respectively and explain how you get it.
- (c) What are the average lengths of the three sets of codewords obtained in (b)?
- (d) Which code is the optimal code for X and how much greater is its expected length than the entropy of X ?